- The recording (1d) is similar to TSA's and the airline's centralized database updated with the traveler's status. In the case of blockchain, the differences are that recording is on a distributed immutable ledger and the use of consensus protocol (1c).

You can think of verification as dealing with general or global requirements within a problem space and validation as being application- or data-specific. In the case of a blockchain application, the transactions are verified and validated according to general rules as well as application-specific rules and conditions.

The bottom two cells (1c and 1d) of the trust quad chart in figure 3.2, consensus and immutable distributed recording, are the responsibility of the blockchain protocol. The purpose of the consensus process is to make sure that a consistent set of transactions (a block) is recorded on the blockchain. As you'll recall from chapter 1, nodes form the blockchain network. An exact copy of each block, chosen by the consensus process, is recorded on all the distributed nodes. The chain of blocks is deemed to be immutable because each of the nodes or stakeholders has a copy; no one node can make changes without its copy going out of sync with the others. The blockchain protocol and infrastructure determine the necessary rules and software for these two cells (consensus and recording) of the trust chart. You can focus on these two aspects of trust when you develop and contribute at the protocol level. As an application developer, you'll design only the application-specific verification and validation (the top two cells, 1a and 1b).

3.1.2 Integrity